



Introduction

Target Book Audience

This book has been written for students interested in finance; traders and quants in hedge funds and pension funds; and investment firms looking to build intuition into when (or when not) and why (or why not) modern trading strategies work well and how to enrich a strategy with artificial intelligence (AI) algorithms to make it more robust.

The book builds upon the following:

- A working knowledge of Python 3.x, including pandas, numpy, and sci-kit libraries.
- Some familiarity with modern finance and trading.
- A reader's constant curiosity to keep asking himself/herself what a change of any part of the algorithm would result in.

Book Goals

We had three principal goals in writing this book:

1. To provide an overview of the key artificial intelligence (AI) algorithms, techniques, and best practices used in modern financial trading,
2. To teach intuition behind the algorithms via a significant number of worked-out real-world examples with detailed explanations, and
3. To provide an easy-to-setup and use environment where readers could instantly experiment with the algorithms to build their confi-

dence without spending any time setting up the required infrastructure.

Unlike other books on financial trading, this book does not focus on setting up market data, backtesting, and trading infrastructure. Such setups are notoriously difficult, expensive, and require major investments. Consider the issue of market data quality, which requires a permanent team of data scientists and software engineers to ensure the data is correct and always delivered without delay.

Instead, the book fully utilizes **QuantConnect**, an open-source, financial-trading-as-infrastructure platform that allows you to start building your trading algorithms and test trading hypotheses using QuantConnect's market data feeds, backtesting, and trading infrastructure.

Similarly, the book does not focus on explaining the intricate details of training large language models or sophisticated financial models. Instead, it depends on **AWS Bedrock**, which has already trained foundation models, or **MLFinLab** or **[PredictNow.ai](#)** trained financial models.

Book Organization

The book is organized as follows:

- In the Preface, we introduce the QuantConnect platform.
- In **[Part I](#)**, we introduce foundational concepts of modern markets (**[Chapter 1](#)**) and quantitative trading (**[Chapter 2](#)**).
- **[Part II](#)** introduces foundational concepts of designing a robust AI algorithm, starting with defining your inputs and output, shaping your data, and choosing the right models to use.
- **[Part III](#)** is the core of the book, describing more than 20 fully implemented algorithms that harness different data sources, models, and technology platforms. Included are algorithms built on

QuantConnect with source code and results, allowing readers to see each algorithm from different perspectives, which is critical for mastering its applications in readers' strategies.

Accompanying Book Materials

Book Examples

The fully worked-out real-world examples have been designed to illustrate key components of modern algorithmic trading strategies.

Their parameters, such as the assets to be traded or the backtesting start and end dates, have been chosen to illustrate the algorithm's performance. Readers are encouraged to modify the input parameters to see the effect of their changes.

The source code of all book examples is available in QuantConnect's library for readers to clone and experiment with. To access them, navigate to hands-on-ai-trading.com/examples, and you'll be directed to complete, maintained examples with which to experiment. To see how to modify an algorithm's parameters, see Parameter Optimization in [Chapter 2](#).

Feedback

For readers with issues regarding bugs in the book code examples, please create an issue in the GitHub repository for the book (qnt.co/book-repo), and the QuantConnect team will address them directly.

For general support regarding issues related to QuantConnect, please log in to the QuantConnect Discord server at quantconnect.com/discord, where the QuantConnect team, AI-quant support agents, and thousands of community members can assist.

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